



MEHRAN UNIVERSITY OF ENGINEERING AND TECHNOLOGY, JAMSHORO
DEPARTMENT OF ELECTRONIC ENGINEERING
INTEGRATED ELECTRONICS # 13



Roll No:	Date:
Checked by:	Score:

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	PRESENTATION	DESIGNING AND CODING	OBSERVATION/ RESULTS	SCORE

Object: To analyze op-amp behavior as an integrator

Apparatus:

- 1) Op-amp 741 IC
- 2) Breadboard
- 3) Few connecting hard wires
- 4) Few connecting leads
- 5) Oscilloscope
- 6) Function Generator
- 7) DC-voltage power supply

Theory:

6.1 Op-amp integrator

An operational amplifier, also known as op-amp is a versatile linear integrated circuit device. This versatile component performs several types of circuit operations.

The ways in which external components are connected to the input and output pins and the kinds of components used determine the type of op-amp circuit.

An integrator operational amplifier is one type of op-amp circuit. The magnitude of its output is determined by the length of time a voltage is present at its input. The output voltage is proportional to the length of time the input is present. The longer the input voltage is present, the greater the output becomes.

The external components connected to an integrator include

- Resistor connected to inverting input
- A capacitor that is used instead of a feedback resistor between the output and the inverting input

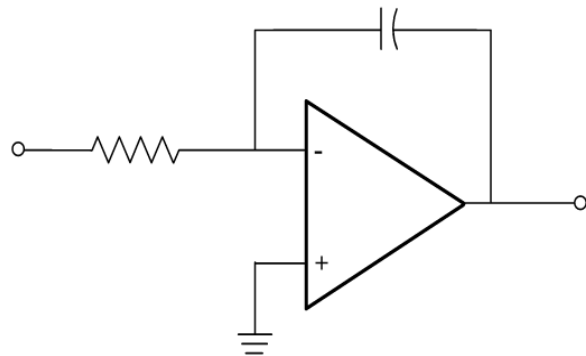


Fig:01 Op-amp Integrator

6.1.1 Passive Integrator

Passive integrator circuit can be designed containing only passive components like capacitor and (or) resistor. There are two possibilities of that discussed below.

✓ Current to voltage C integrator

What is the most elementary electrical integrator? Of course, this is the humble capacitor. So, if we drive a capacitor C by a constant current source I , it acts as an ideal current-to-voltage integrator with a current input I_{IN} and a voltage output $V_{OUT} = V_C$. Note that the output voltage changes linearly through the time.

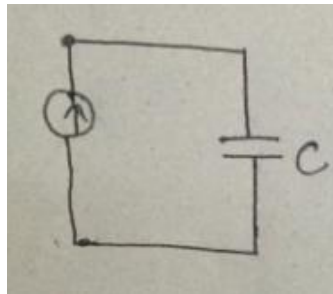


Fig:02 Simplest I-V converter

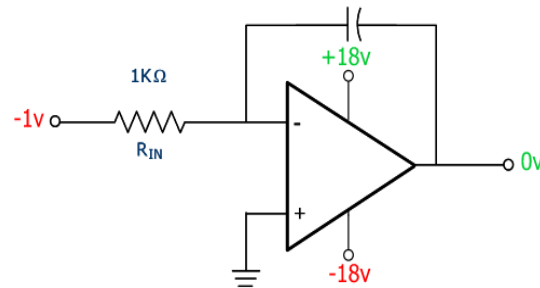
✓ Voltage to voltage RC Integrator

✓ How it works

When the voltage is first applied to an integrator op-amp

- The uncharged capacitor acts like a short and has very little resistance

- Resistance ratio of the capacitor/input resistor is very low.
- Gain of op-amp is less than 1
- The output voltage is 0 volts.



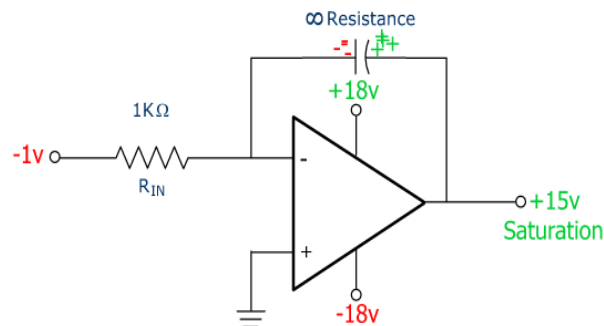
Fig;05 Integrator at start up

As capacitor is charging

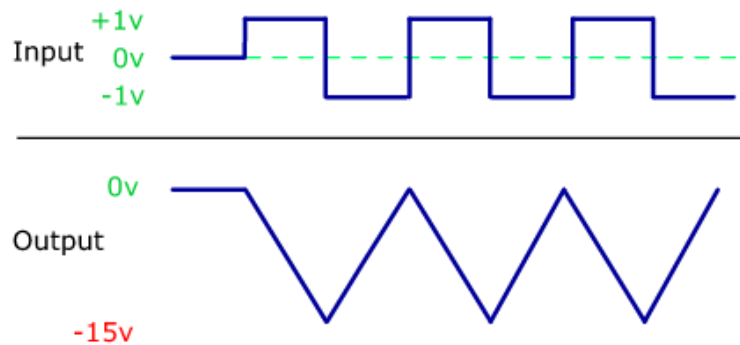
- Effective resistance of capacitor is increasing
- Resistance ratio of capacitor/input resistor is increasing
- Voltage at output is increasing

When the capacitor is fully charged

- Capacitor acts like open with resistance
- Resistance ratio of capacitor/input resistor is infinite
- Gain is infinite, causing the op-amp to go into positive saturation



Fig;06 Op-amp Integrator-capacitor fully charged



Fig; 07 Op-amp Integrator input/output curves

Short answer questions

1. Enlist few applications of op-amp integrator
 - Analog Computers
 - Analog-to-digital converters
 - Wave-shaping circuits
2. Differentiate between active and passive integrators.

A passive integrator is a circuit which does not use any active devices like op-amp or transistors. An integrator circuit which consist of active devices is called an Active integrator provides a much lower output resistance and higher output voltage than is possible with a simple RC circuit.

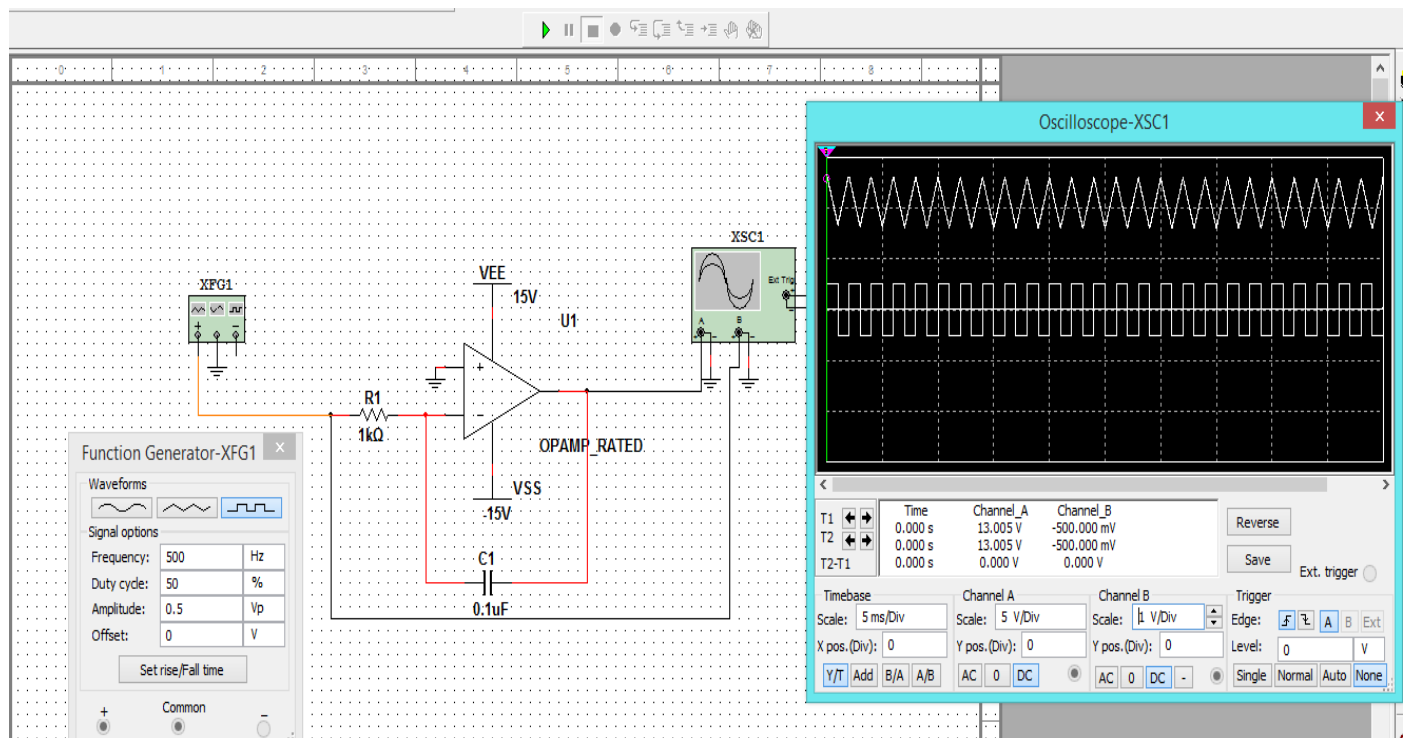
Lab Activity

1. Design an op-amp based circuit whose output voltage is something like the following equation; attach the waveform of V_0 with this handout.

$$V_0 = \frac{1}{1 \times 10^{-4}} \int 0.5 dt$$

Op-amp Integrator circuit

$V_{in} = 0.5$, feedback capacitor = 0.1uF, $R_{in} = 1\text{kohm}$



ADD MORE CIRCUITS ALONG WITH RESULTS...